

# Rethinking Reconstruction-Based Jet Anomaly Detection

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# Introduction

## Rethinking Reconstruction-Based Jet Anomaly Detection

We study QCD (background) vs WJet (signal) anomaly detection under a background-only reconstruction setting. A supervised QCD/WJet classifier is trained as an upper-bound baseline, while the main unsupervised approach trains autoencoders only on QCD and uses reconstruction loss as the anomaly score. We improved the data processing pipeline, added plotting utilities, refined the GAE bottleneck design, and explored several Transformer variants, including masked token reconstruction, CLS-conditioned prototype reconstruction with Hungarian matching, and position-conditioned token-to-token reconstruction.

Across these studies, we find that reconstruction-based anomaly detection is highly sensitive to model capacity, inductive bias, and decoder design. Smaller and more constrained models often perform better, while more expressive Transformer autoencoders can learn reconstruction rules shared by QCD and WJet, causing AUC to degrade toward 0.5. Latent-space visualizations show strong QCD/WJet overlap, suggesting that current models do not learn directly separable representations and rely mainly on decoder residuals. This motivates a shift from reconstruction loss as a byproduct toward representation learning objectives that encode distribution-level differences more explicitly.

# Modeling Assumption

## Underlying Reconstruction-Based Detection

The reconstruction-based anomaly detection relies on the core assumption that:

If the model is only trained on background jets, and the backgrounds behave differently than the signal jets, then the model will perform worse in reconstructing the signals.

However, this property requires the following to be true: (1) limited model capacity and (2) strong architectural bias. Further, this property is not directly optimized by the training objective. The model is only trained to reconstruct its input; anomaly sensitivity appears only as a byproduct of (1) and (2), and the choice of reconstruction target. A lower reconstruction loss does not necessarily imply a better anomaly detector.

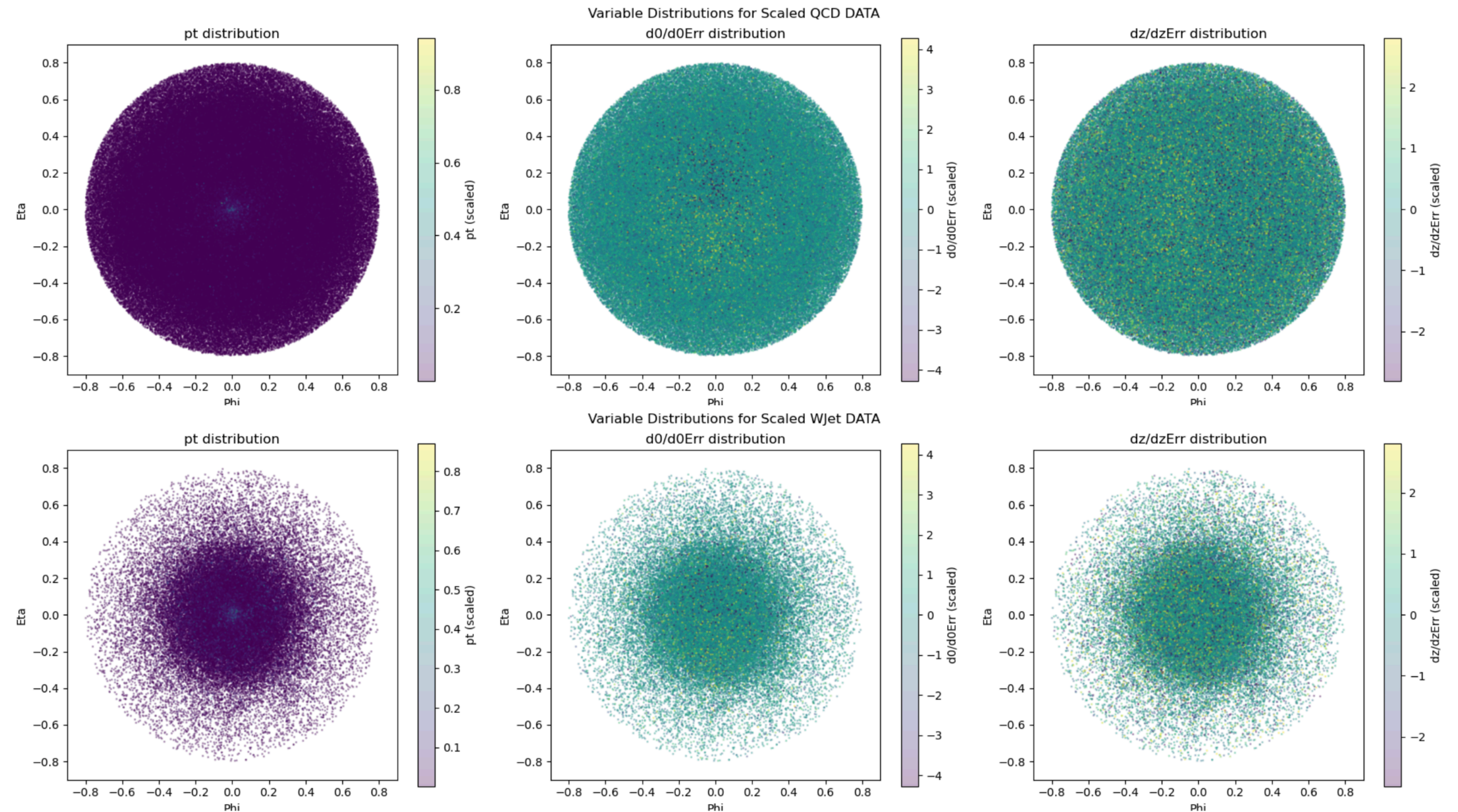


# Dataset

## QCD and WJet

Our dataset includes 125,103 QCD (background) jets and 4,067 WJet (signal) jets. They are selected from a broader dataset by limiting  $p_t$  to be between 200 and 400.

Right: All events plotted together. Each point represents a PFCand. Features are normalized.



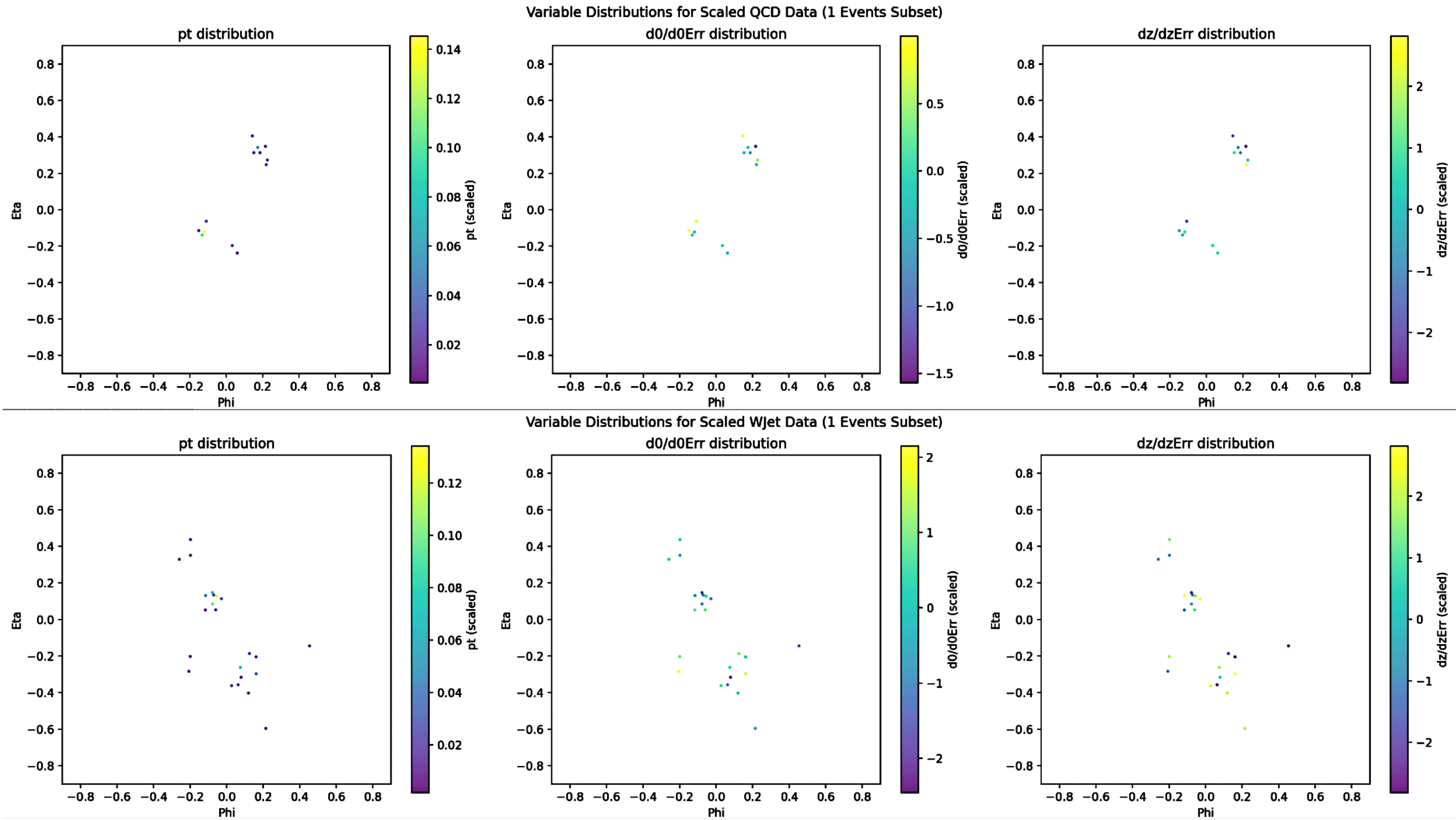


# Dataset

## QCD and WJet

Right: A randomly selected single event from QCD and WJet respectively.

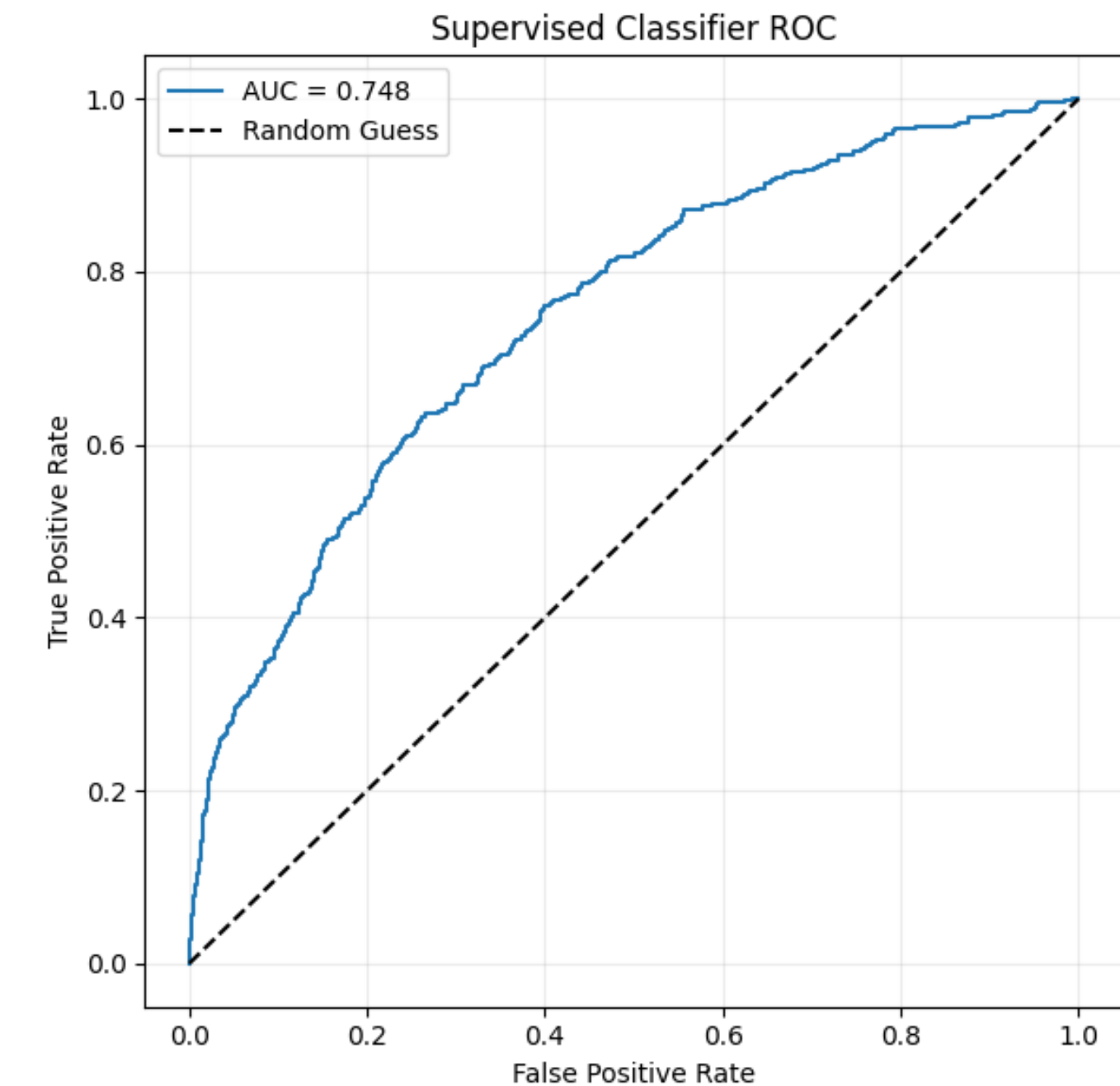
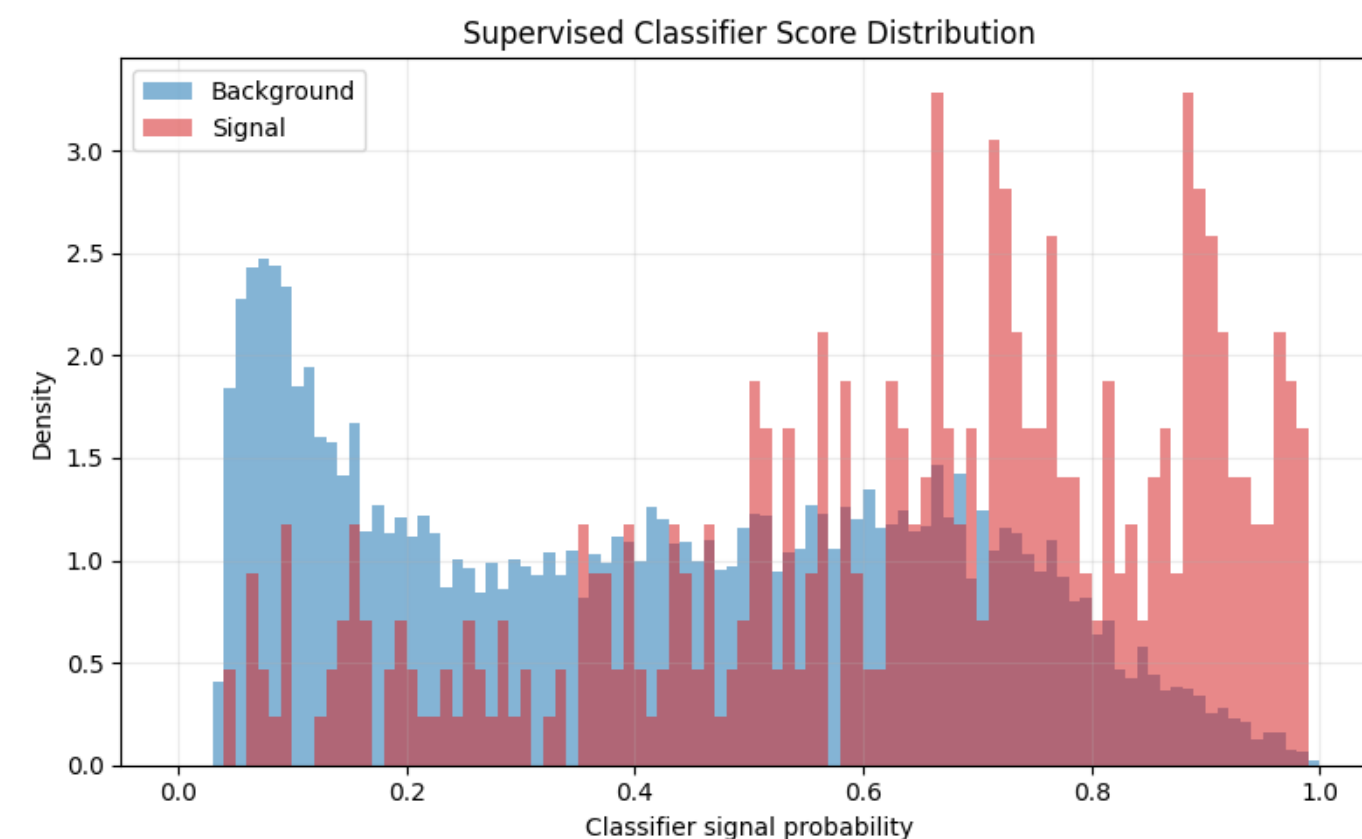
(Image enhanced for visibility)



# Classifier Upper Bound

## Supervised Classifier Trained on QCD vs WJet

The graph neural network (GNN) supervised classifier is trained to explicitly classify QCD and WJet events. It indicates the amount of information available for discerning two distributions in our dataset, and serves as an upper bound on how well any unsupervised model might perform.



```
Train split: bg=74785 signal=3406 used=6812
Validation split: bg=9348 signal=426
Test split: bg=9348 signal=426
Model Summary:
JetGraphAutoencoderClassification(
  (conv1): EdgeConv(nn=Sequential(
    (0): Linear(in_features=32, out_features=128, bias=True)
    (1): ReLU()
    (2): Linear(in_features=128, out_features=128, bias=True)
  ))
  (conv2): EdgeConv(nn=Sequential(
    (0): Linear(in_features=256, out_features=64, bias=True)
    (1): ReLU()
    (2): Linear(in_features=64, out_features=64, bias=True)
  ))
  (conv3): EdgeConv(nn=Sequential(
    (0): Linear(in_features=128, out_features=64, bias=True)
    (1): ReLU()
    (2): Linear(in_features=64, out_features=64, bias=True)
  ))
  (conv4): EdgeConv(nn=Sequential(
    (0): Linear(in_features=128, out_features=128, bias=True)
    (1): ReLU()
    (2): Linear(in_features=128, out_features=128, bias=True)
  ))
  (fc3): Linear(in_features=128, out_features=32, bias=True)
  (out): Linear(in_features=32, out_features=1, bias=True)
)
Number of trainable parameters: 90945
```

# Revisiting GAE

## Past GAE Baseline

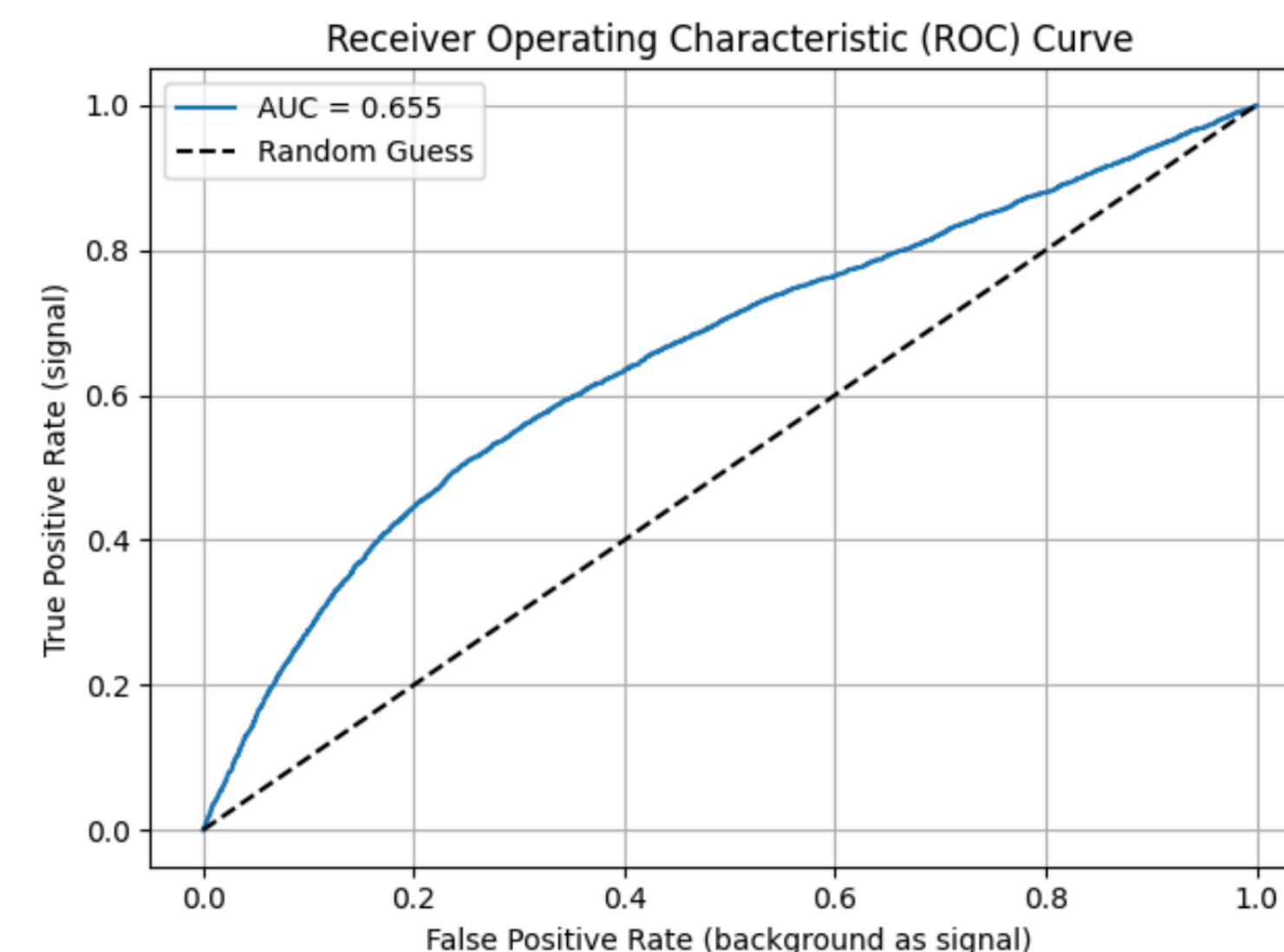
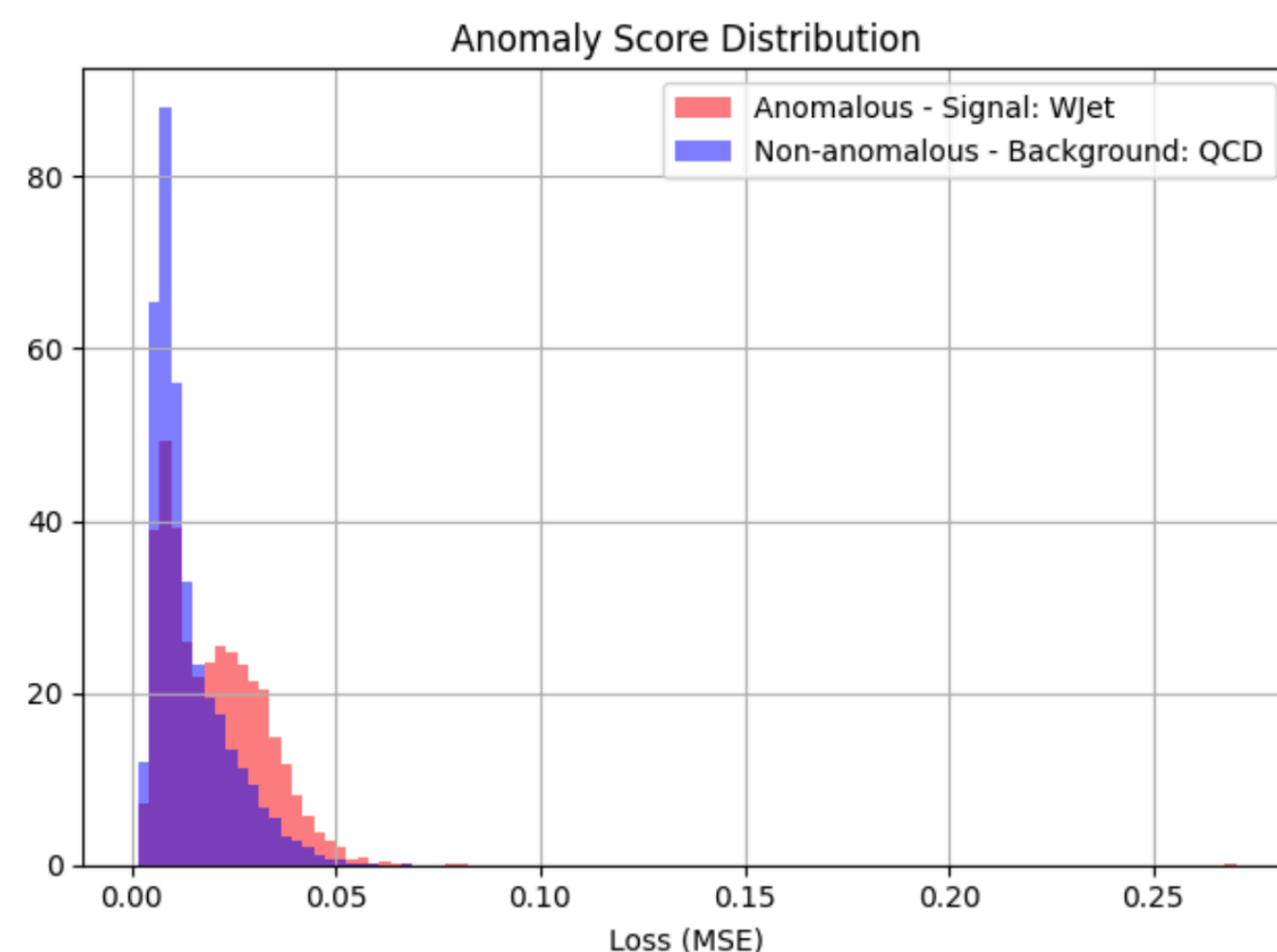
We first revisited the graph autoencoder (GAE) baseline.

The GAE takes in a graph, where each node represents a particle, and edges are constructed using kNN on hand-crafted node features. There is no edge feature. Each node has features including **[eta, phi, pt, d0/d0Err, dz/dzErr]**. Node features are normalized according to the pipeline from last year. (We speed up data preprocessing by 30x by resolving several bottlenecks.)

The original GAE architecture had a weak bottleneck, in some cases even larger than the input feature dimension, which made the reconstruction task insufficiently constrained.

With 20 epochs, feature\_dim = 5, hidden\_dim = 32 and latent\_dim = 16, the GAE baseline achieved 0.655 AUC

```
INFO:root:Model Summary:
JetGraphAutoencoder(
  (conv1): EdgeConv(nn=Sequential(
    (0): Linear(in_features=10, out_features=32, bias=True)
    (1): ReLU()
    (2): Linear(in_features=32, out_features=32, bias=True)
  ))
  (conv2): EdgeConv(nn=Sequential(
    (0): Linear(in_features=64, out_features=16, bias=True)
    (1): ReLU()
    (2): Linear(in_features=16, out_features=32, bias=True)
  ))
  (conv3): EdgeConv(nn=Sequential(
    (0): Linear(in_features=64, out_features=16, bias=True)
    (1): ReLU()
    (2): Linear(in_features=16, out_features=16, bias=True)
  ))
  (conv4): EdgeConv(nn=Sequential(
    (0): Linear(in_features=32, out_features=32, bias=True)
    (1): ReLU()
    (2): Linear(in_features=32, out_features=5, bias=True)
  ))
)
INFO:root:Number of trainable parameters: 5525
```





# Revisiting GAE

## What we did

We improved the GAE by:

- A. Decoupling hidden dimension with feature dim and latent dim.  
Previous baseline had them hard-coded.
- B. Limiting the latent dim to be smaller than feature dim, creating effective information bottleneck.
- C. Using a Transformer-FFN style scaling in EdgeConv Linear layers.

Right: Example improved GAE architecture with 5 input feature dim, 8 hidden dim, and 2 latent dim. Additional embedding and unembedding layers decouple feature dim from hidden dim.

```
Model Summary:
JetGraphAutoencoder(
  (embedding): Linear(in_features=5, out_features=8, bias=True)
  (unembedding): Linear(in_features=8, out_features=5, bias=True)
  (conv1): EdgeConv(nn=Sequential(
    (0): Linear(in_features=16, out_features=64, bias=True)
    (1): ReLU()
    (2): Linear(in_features=64, out_features=8, bias=True)
  ))
  (conv2): EdgeConv(nn=Sequential(
    (0): Linear(in_features=16, out_features=64, bias=True)
    (1): ReLU()
    (2): Linear(in_features=64, out_features=2, bias=True)
  ))
  (conv3): EdgeConv(nn=Sequential(
    (0): Linear(in_features=4, out_features=64, bias=True)
    (1): ReLU()
    (2): Linear(in_features=64, out_features=8, bias=True)
  ))
  (conv4): EdgeConv(nn=Sequential(
    (0): Linear(in_features=16, out_features=64, bias=True)
    (1): ReLU()
    (2): Linear(in_features=64, out_features=8, bias=True)
  ))
)
Number of trainable parameters: 5367
```



# Revisiting GAE

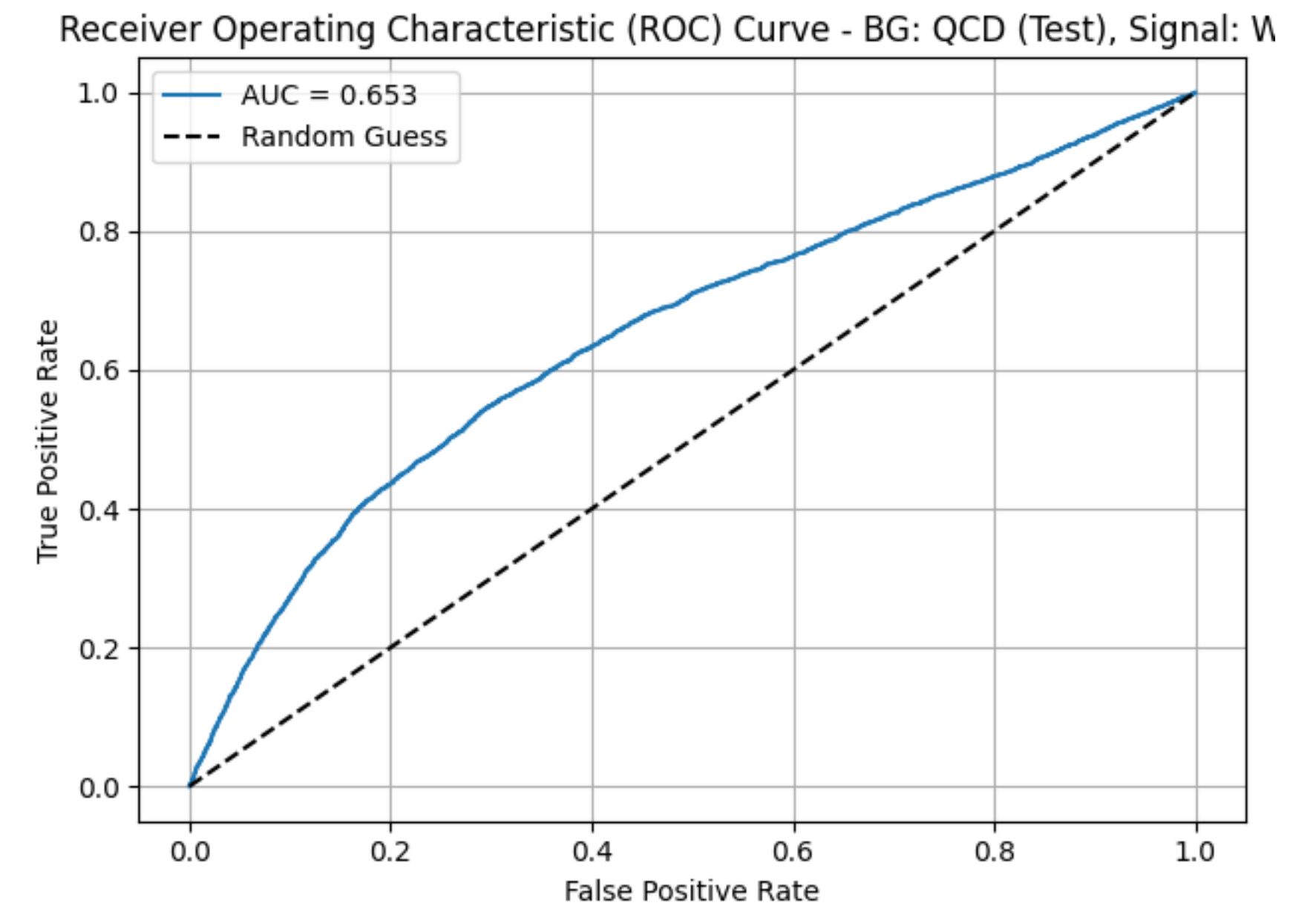
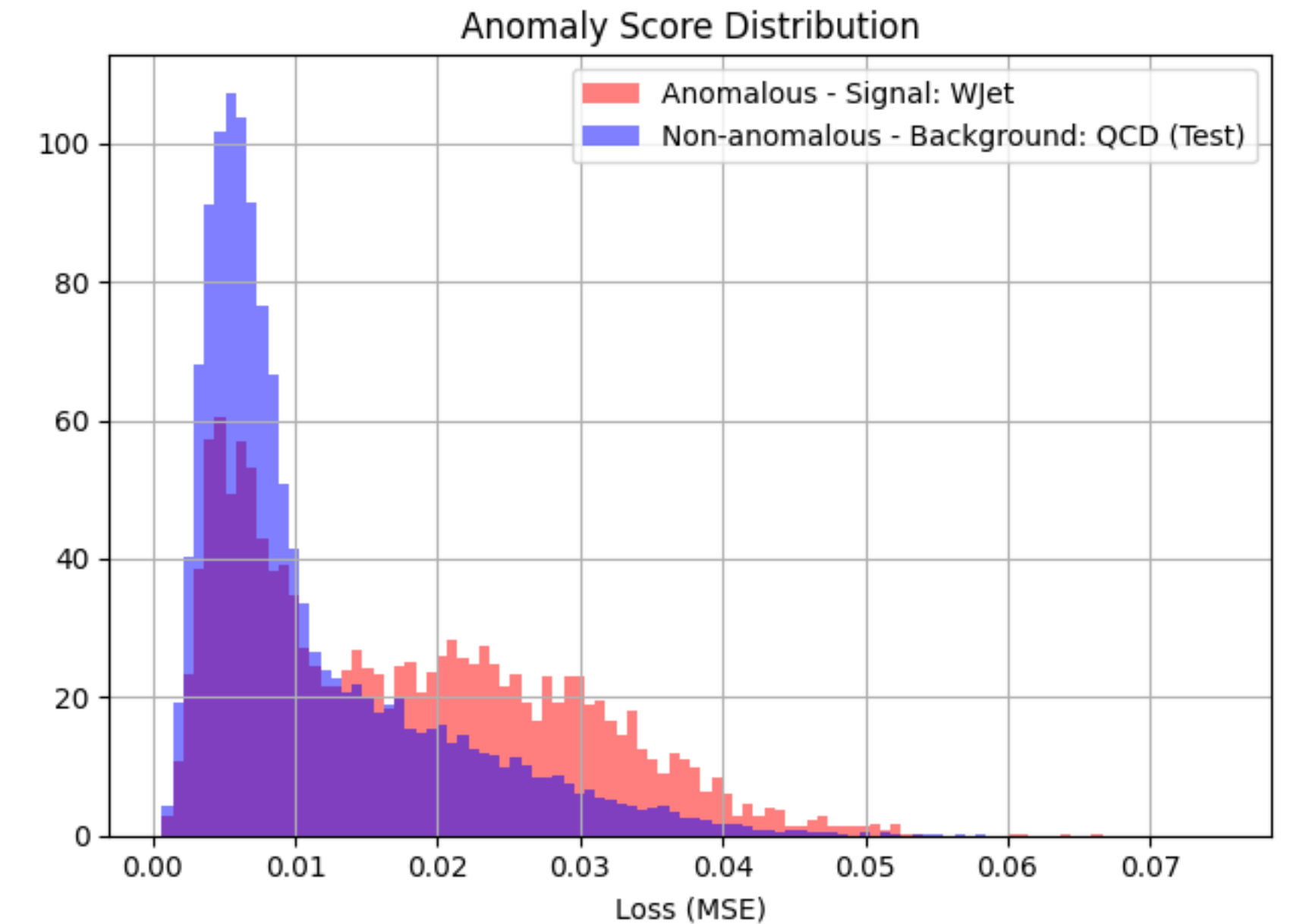
## What we did

After improving the bottleneck design, the GAE produced more stable anomaly detection performance. The GAE baseline, with insufficiently constrained information bottleneck, could learn identity mapping in long training. The improved GAE guarantees this will not happen.

We conducted hyper-parameter scans covering:

- (1) hidden dim from 8 to 64
- (2) latent dim from 2 to 16
- (3) kNN edge count 3 vs 16
- (4) number of encoder and decoder layers from 1 to 8
- (5) adding additional node features, such as charge and mass

The best performing one has 8 hidden dim, 4 latent dim, 3 kNN edge count, and 2 encoder and decoder layers. It achieves 0.653 AUC, which is similar to the GAE baseline's 0.655 within experiment error.



# Revisiting GAE

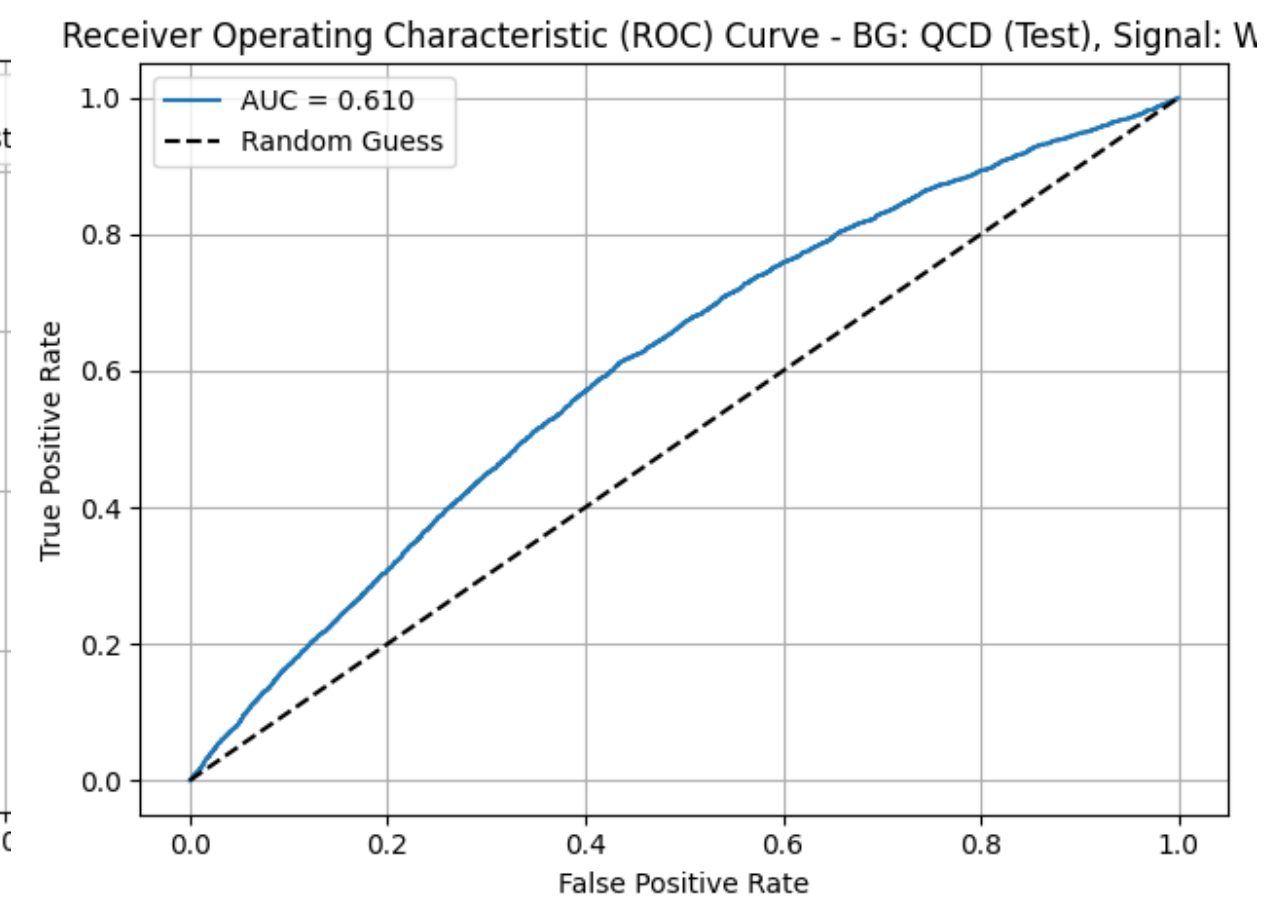
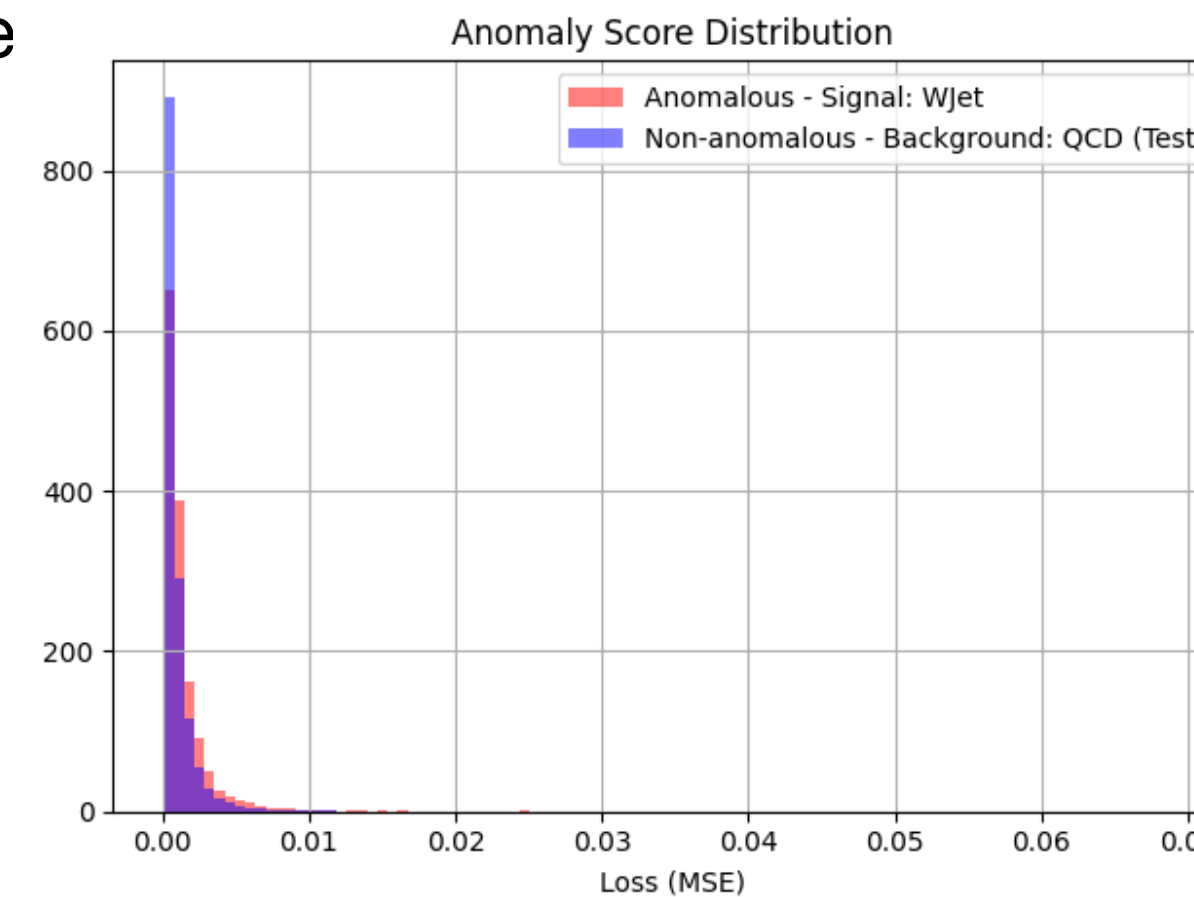
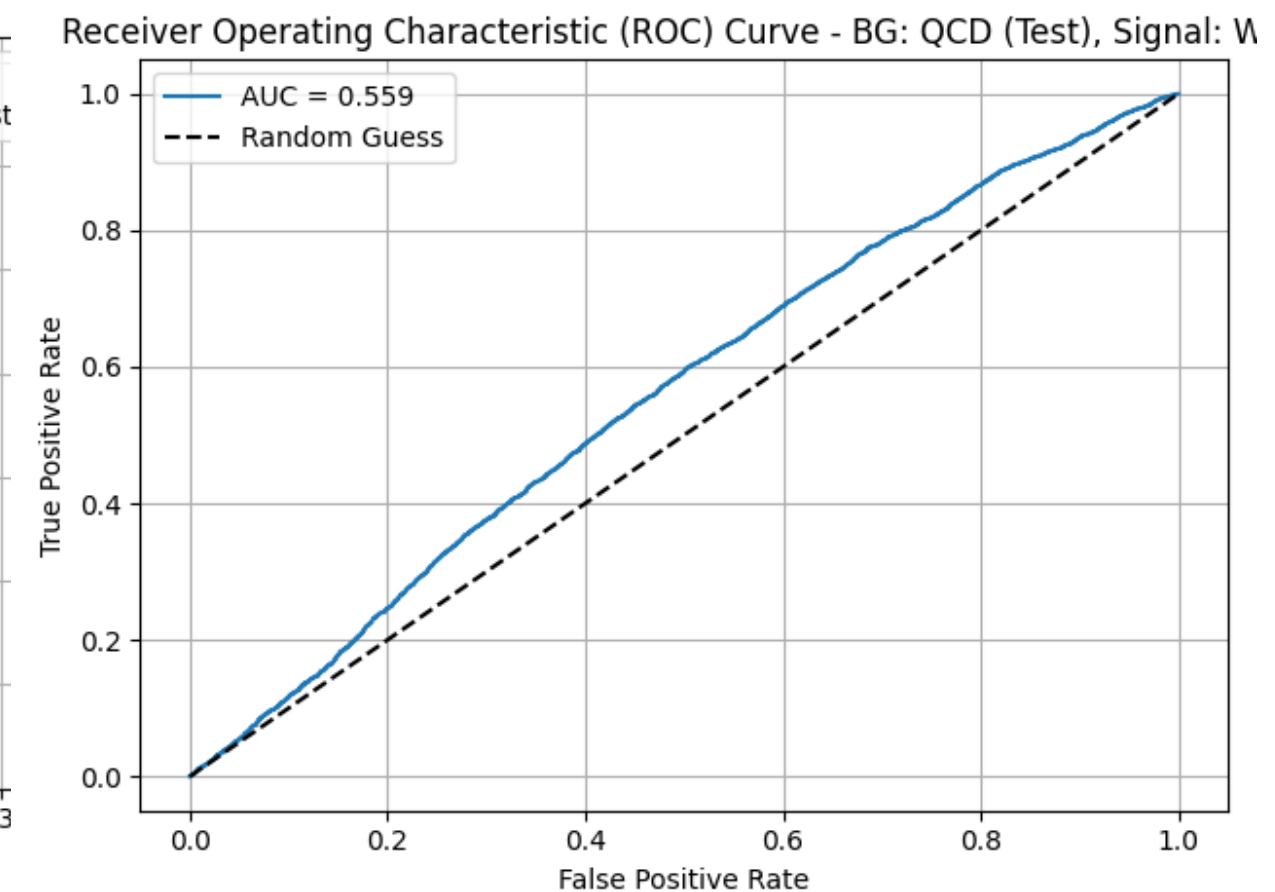
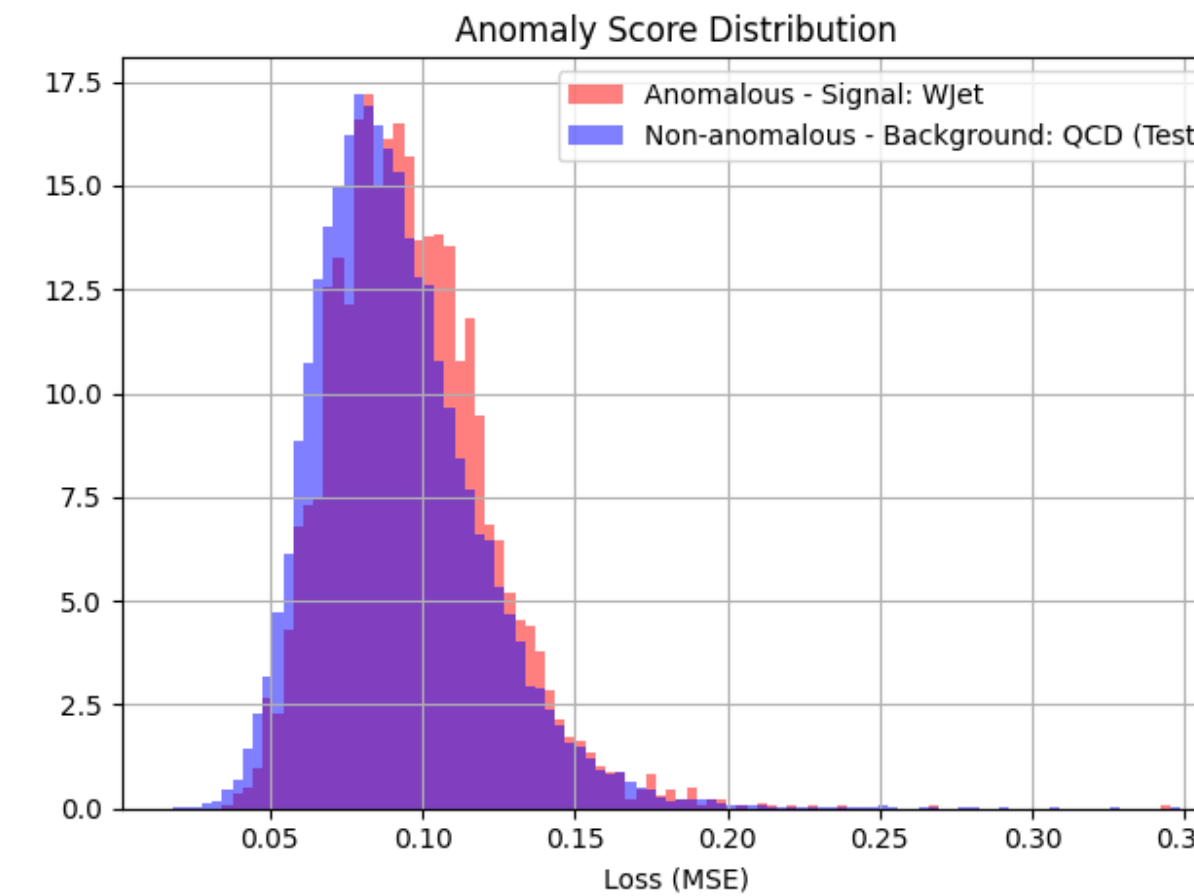
## ROC Performance Volatile Depending on Hyper-params

If GAE capacity is too small, it would be equally bad at reconstructing QCD and WJet. (Top plots: GAE with 1 encoding and decoding layer)

If GAE capacity is too big, it easily learned semi-identity mapping. (Bottom plots: GAE with 32 hidden dim and 16 latent dim, but we only have 5 feature dim)

ROC performance is closer to random guessing in both cases.

This suggests that the GAE works partly because of its graph inductive bias and partly because its limited capacity prevents it from reconstructing unseen events equally well. This raises a limitation: the graph is constructed from hand-chosen features, and the success of the method depends on human-designed structure and a carefully restricted bottleneck.



# Transformer Models

## A Step To Less Human-factor And Inter-particle Relation Modeling

In the GAE baseline, the model compresses each node feature vector independently into a lower-dimensional representation, but it does not compress the number of nodes. If the original particle features are already low-rank, a per-node bottleneck may still reconstruct well without forcing the model to understand inter-particle relationships. To avoid this, we borrow the power of CLS token from Transformer models to make use of global representation

Additionally, the presence of hand-crafted edge construction motivated us to make use of Transformer's attention mechanism. Transformer can be viewed as a graph neural network operating on a fully-connected graph with dynamic edge weights. Since GAE did not create, learn, or use edge features separate from node features, it is better to give this degree of freedom to the model.

The hope was to reduce the dependence on manually constructed graphs and allow the model to learn particle relations directly from the event. We tested several variants, including (1) BERT-style masked token reconstruction, (2) CLS-conditioned prototype reconstruction with Hungarian matching, and (3) position-conditioned token-to-token reconstruction. These architectures probe different ways of turning a particle set into a reconstruction task.



# Transformer Models

## A Step To Less Human-factor

For all Transformer models, we assume the following:

- (1) Each particle node is treated as a token
- (2) Each jet event is treated as a token sequence
- (3) A CLS token is always prepended to input token sequence
- (4) Position encoding by eta-phi

# BERT-style Masked Token Reconstruction

"Fill-in the blank"

A Transformer encoder receives the full event with a random subset of particle tokens masked. The unmasked particles provide the event context, while the masked particles are replaced by a learned mask token added with its position encoding.

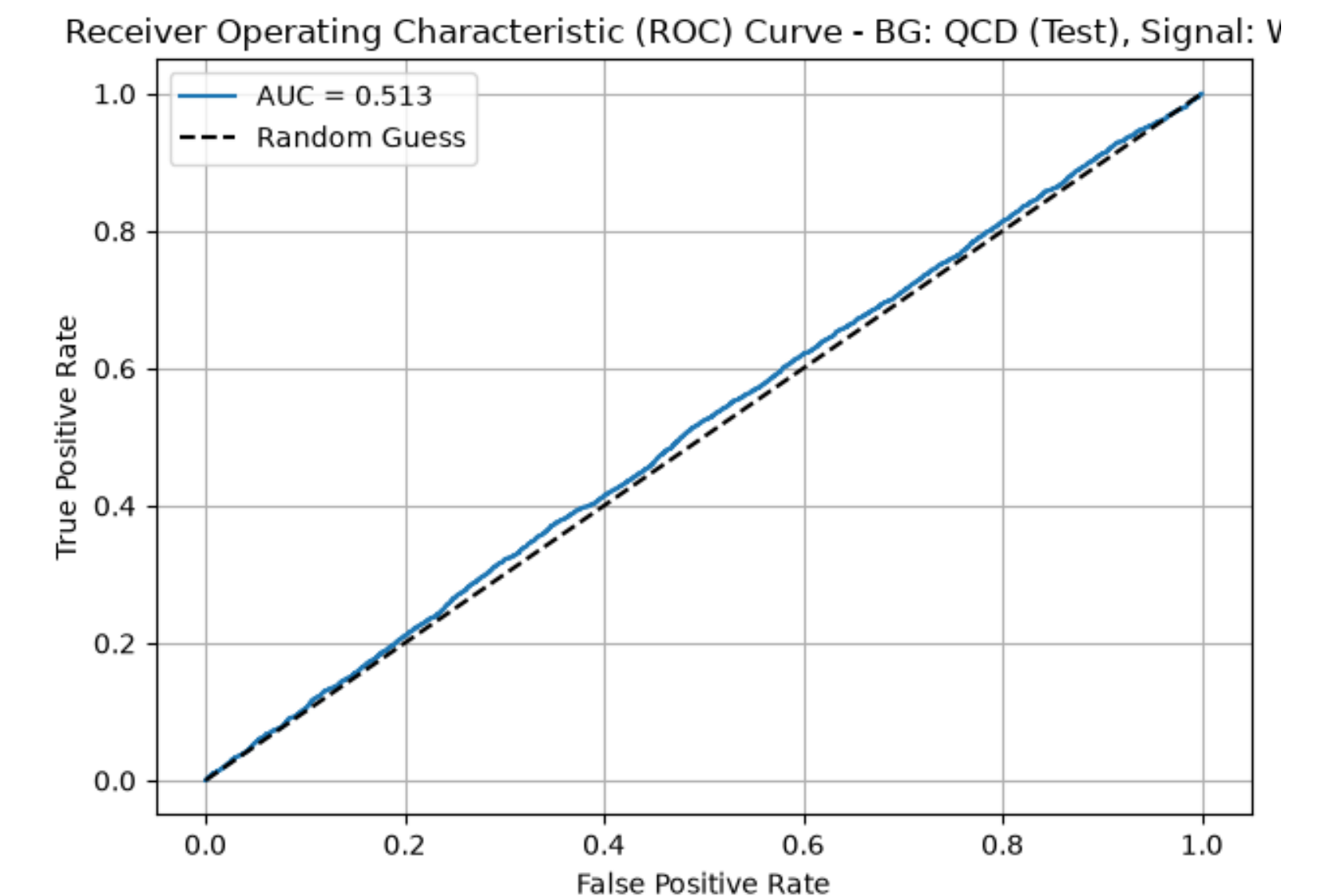
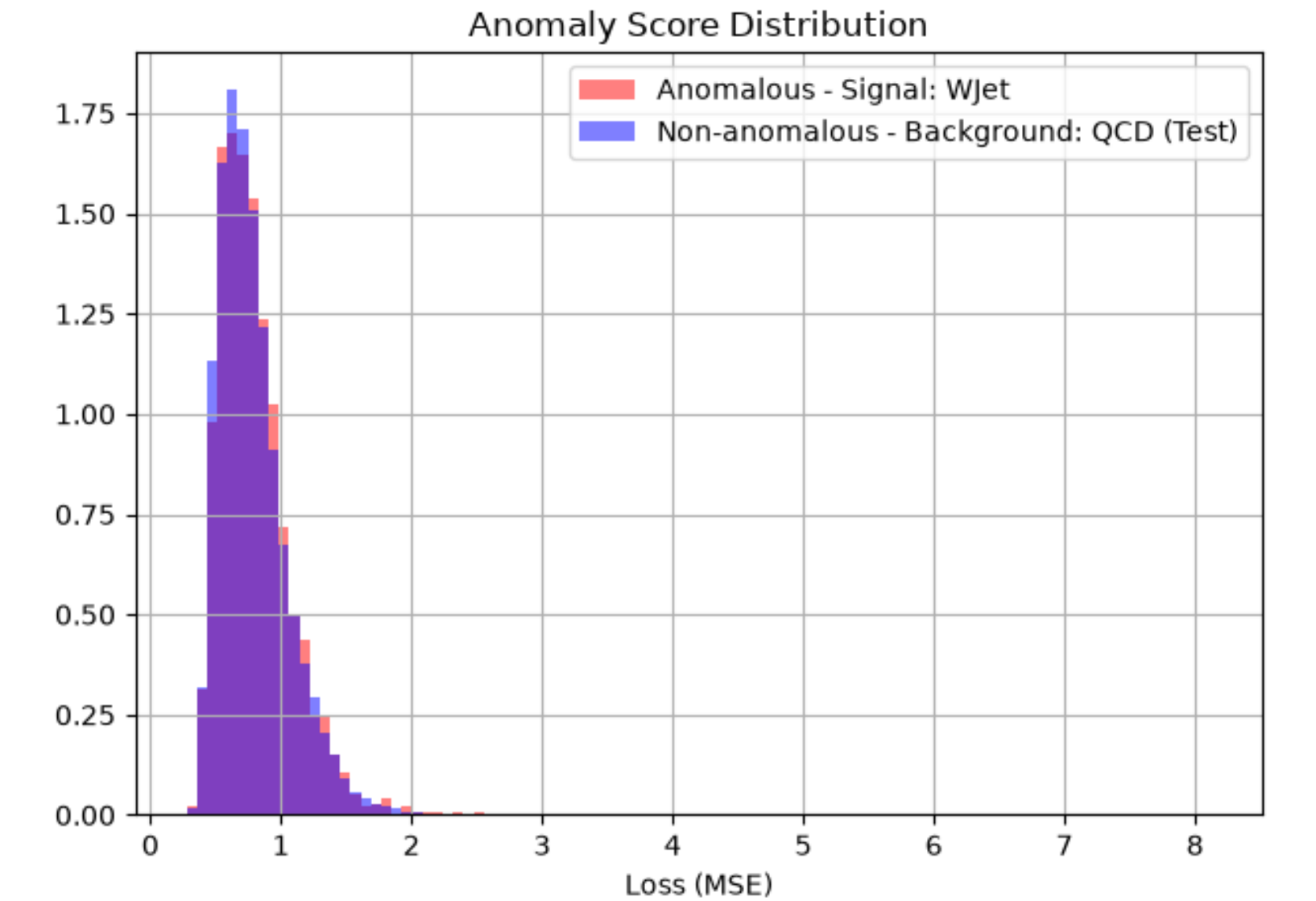
The training target is to reconstruct only the masked particle features from the surrounding event context. The model is trained on QCD events with MSE loss evaluated on masked tokens only.

At evaluation time, the masked-token reconstruction loss is used as the anomaly score: if WJet events follow different local particle-context rules from QCD, their masked particles should be harder to predict and produce higher reconstruction loss.

# BERT-style Masked Token Reconstruction

"Fill-in the blank"

The masked-token reconstruction target failed almost completely as an anomaly score, with ROC AUC close to 0.5. This suggests that, given the surrounding event context, the rule for estimating a masked particle feature is largely shared between QCD and WJet. In other words, masked particle prediction does not necessarily learn the features that distinguish the two event distributions.





# CLS-conditioned Prototype Reconstruction

## Forcing Global Representation

GAE compresses per-node features but keeps the number of nodes unchanged. If particle features are already low-rank, this bottleneck may not force the model to learn inter-particle relationships. We therefore use a Transformer encoder to compress the entire event into one fixed-size CLS latent vector, so the decoder must reconstruct the particle set from event-level information only.

Forward pass:

- (1) Particle features are embedded and combined with eta-phi positional embeddings
- (2) A learnable CLS token is prepended
- (3) The Transformer encoder processes the full event, only the final CLS token is projected into the latent space.
- (4) The decoder receives no original particle tokens. Instead, it starts from learned output slots/prototypes, conditions them on the CLS latent vector, and outputs a reconstructed unordered particle set.

Training target:

Because output slots and target particles have no predefined ordering, we use Hungarian matching to assign reconstructed slots to true particles with minimum total MSE cost. The loss is the average MSE over matched pairs. The anomaly score is the event-level reconstruction loss.

# CLS-conditioned Prototype Reconstruction

## Forcing Global Representation

We conducted hyper-parameter scan:

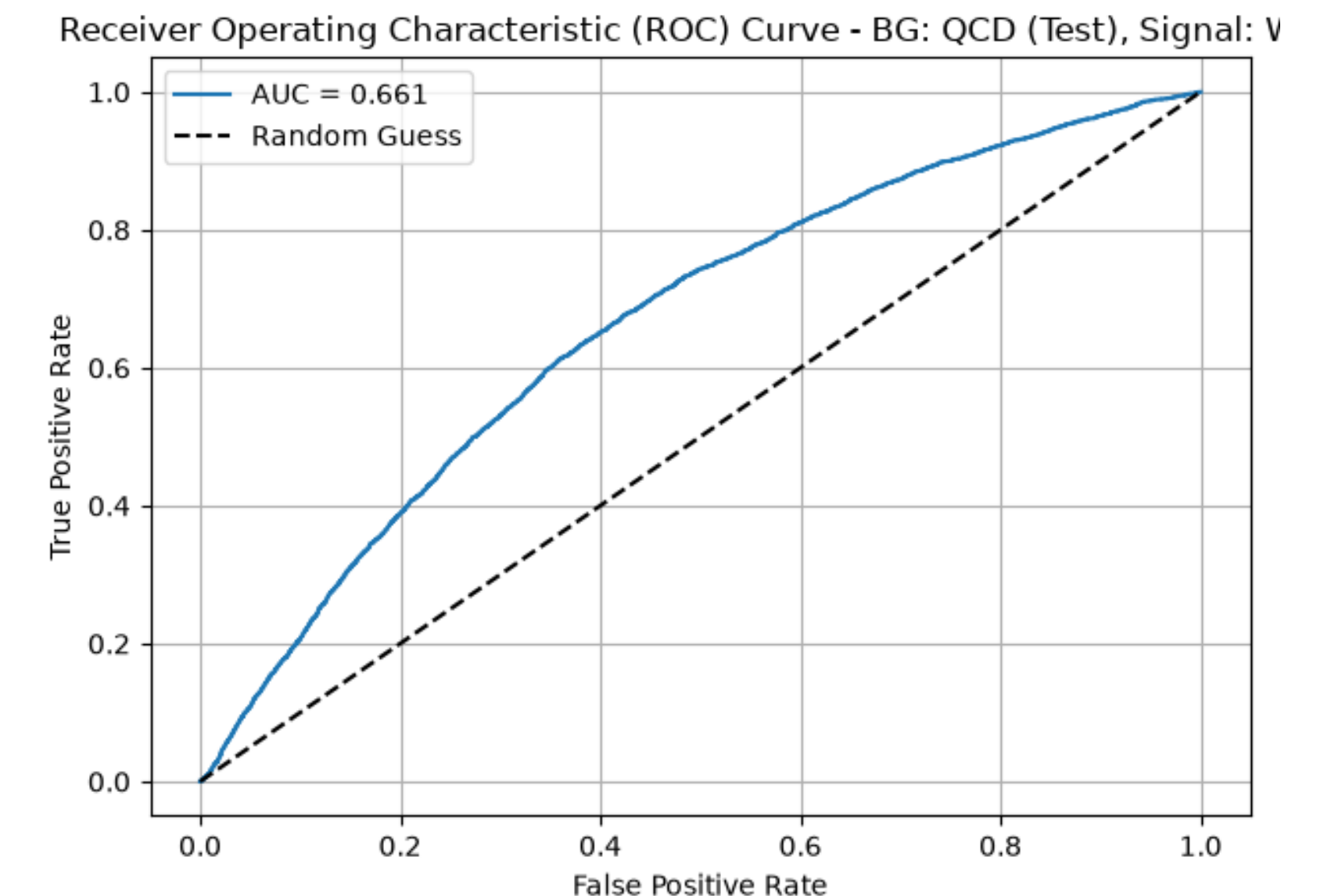
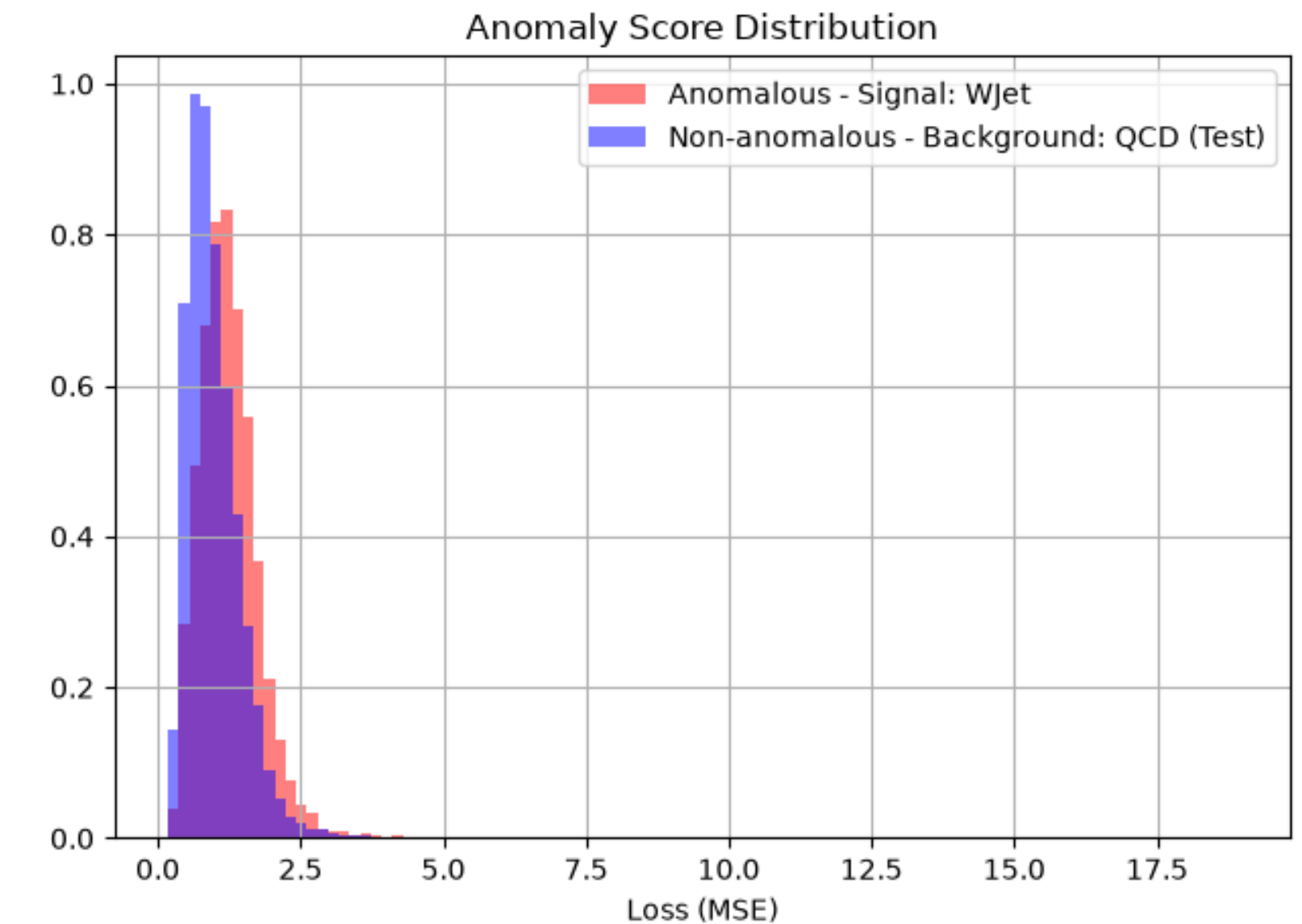
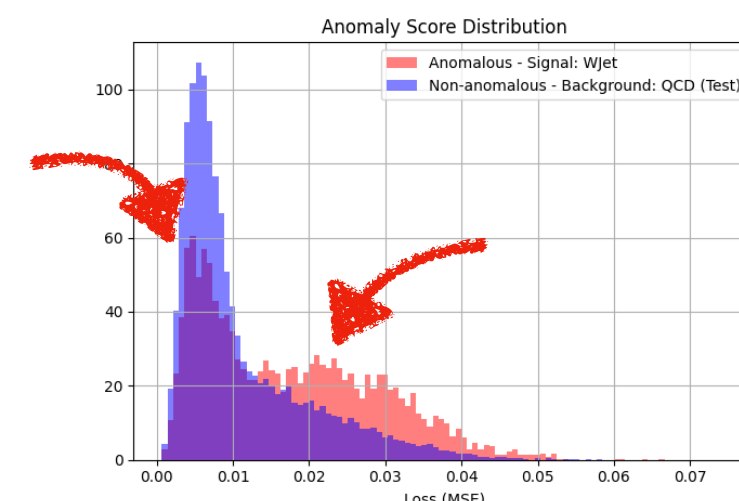
- (1) hidden dim from 2 to 128
- (2) latent dim from 2 to 64
- (3) number of layers from 1 to 4
- (4) training with different epochs

The best model achieves 0.661 AUC with hidden dim 4, latent dim 2, 4 encoder and decoder layers, and 10 epochs.

This is current SOTA, compared to 0.655 GAE highest.

Right: anomaly score distribution and ROC curve for this model.

Note that the bimodal distribution in anomaly's anomalous score (red) is gone. Every GAE exhibits the bimodal pattern.

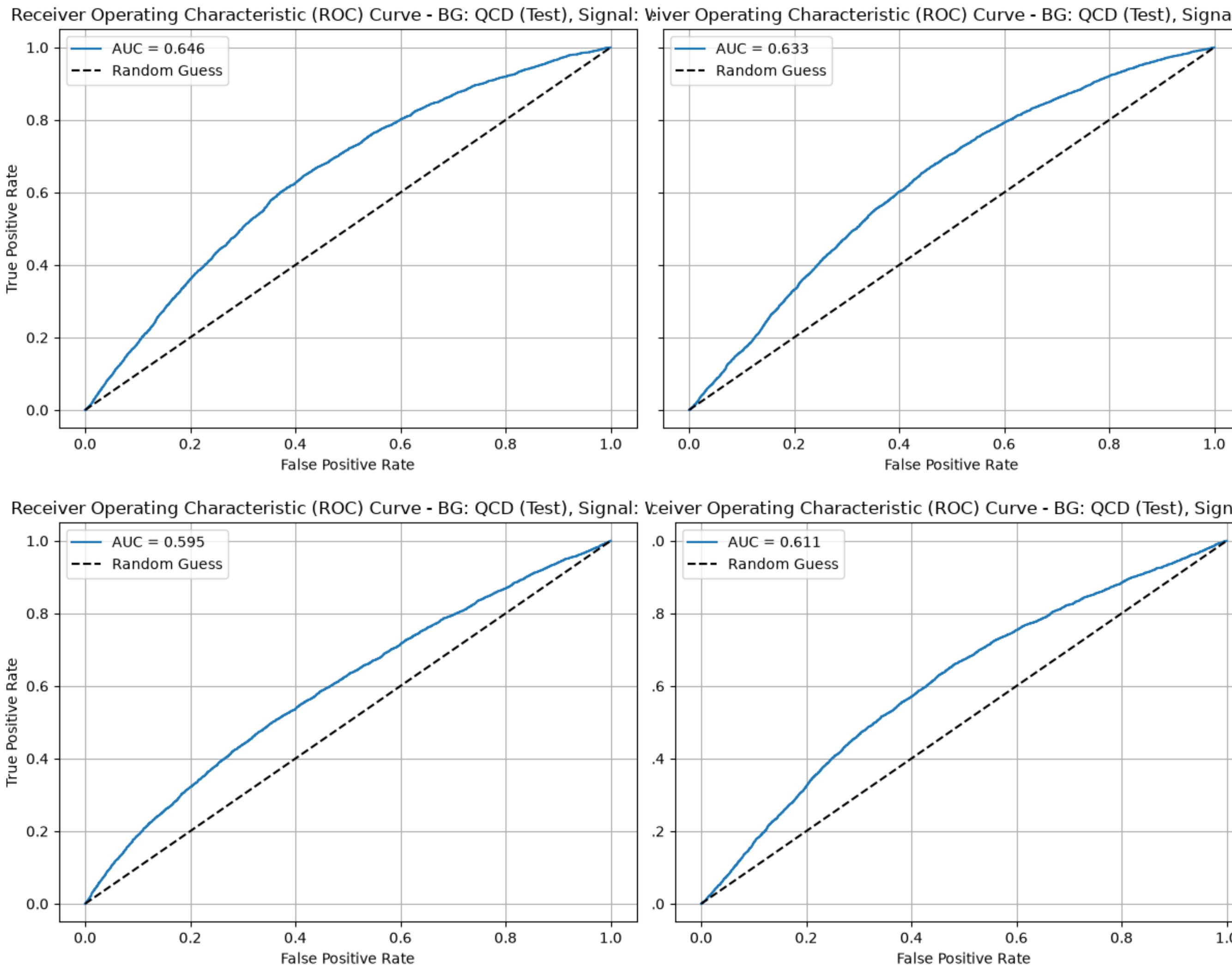


# CLS-conditioned Prototype Reconstruction

## ROC Performance Volatile Depending on Hyper-params

This Transformer AE does not mediate the problem of sensitive model capacity underlying reconstruction-based anomaly detection, although the ROC performance is steady across a wider range of model sizes.

- Top left: 4 hidden, 2 latent, 1 layer, AUC = 0.646
- Top right: 8 hidden, 2 latent, 2 layers, AUC = 0.633
- Bottom left: 32 hidden, 8 latent, 4 layers, AUC = 0.595
- Bottom right: 64 hidden, 16 latent, 4 layers, AUC = 0.611





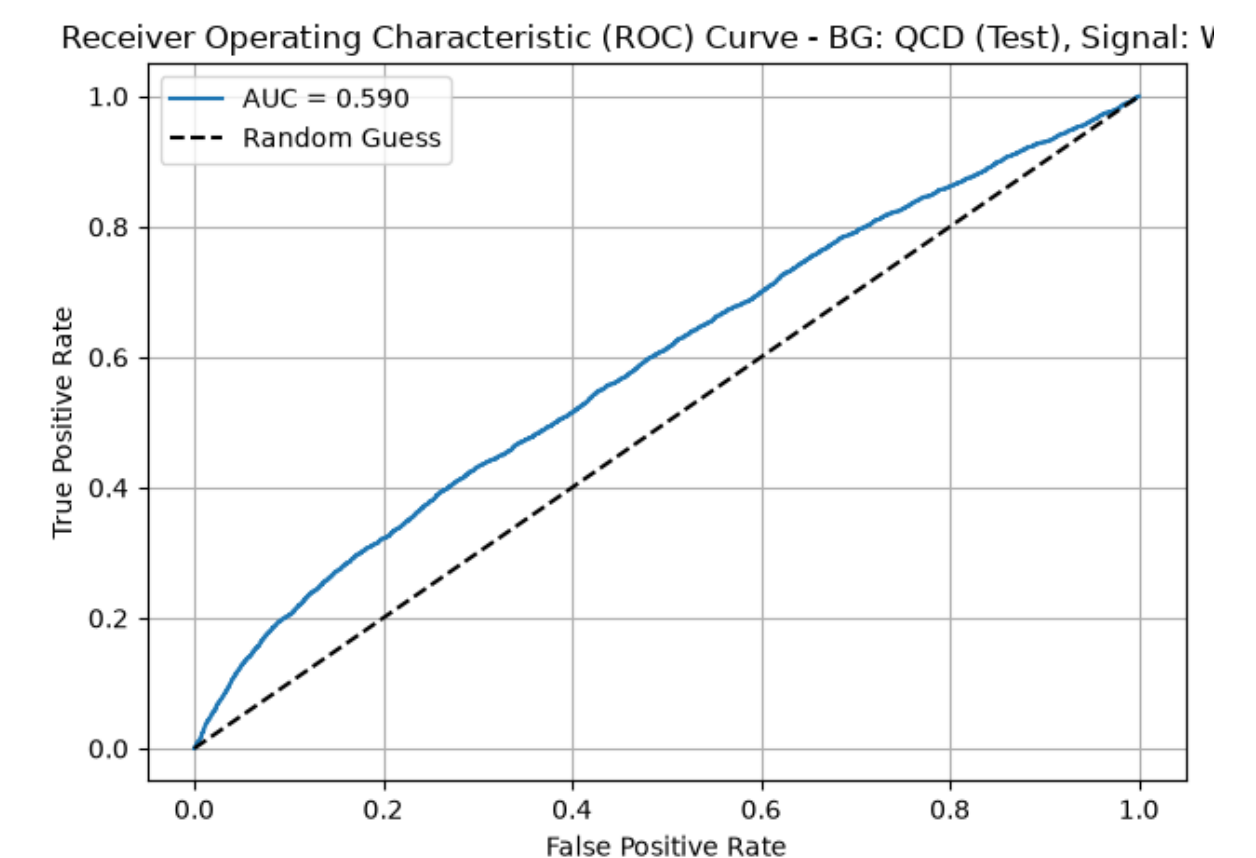
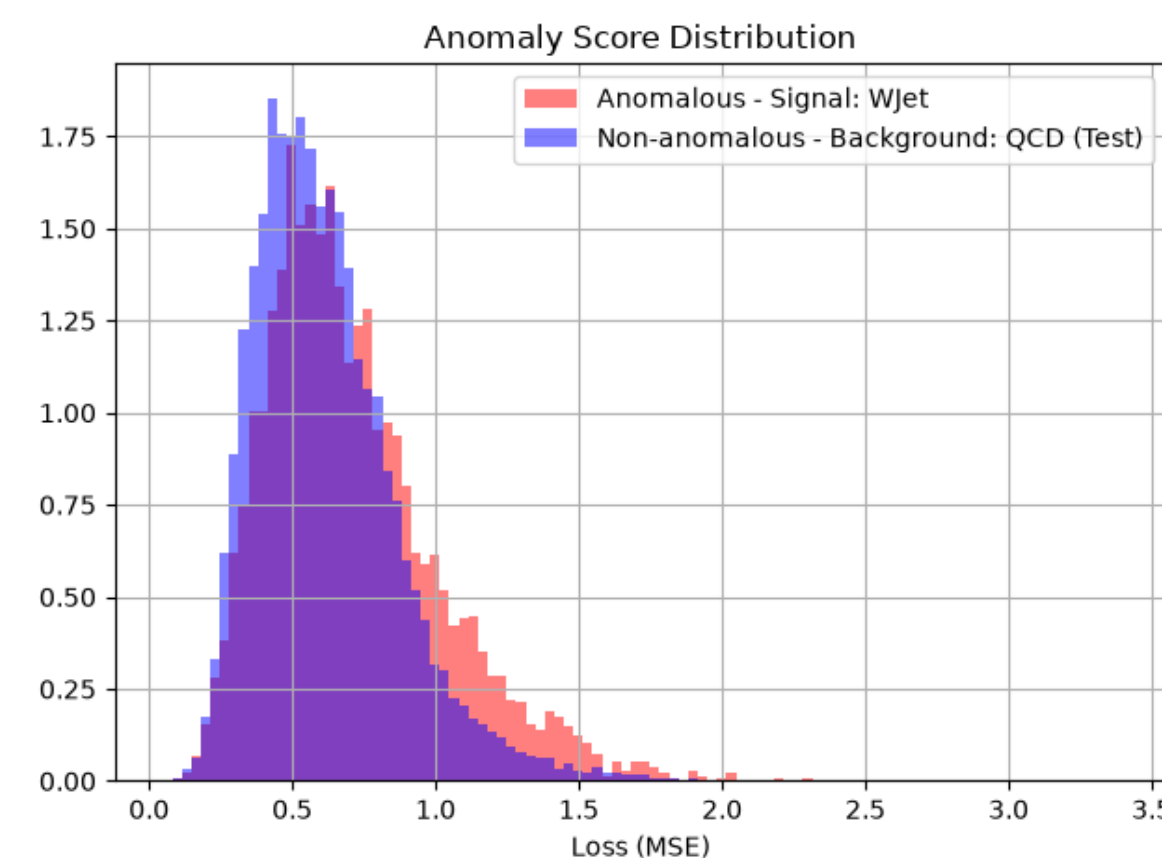
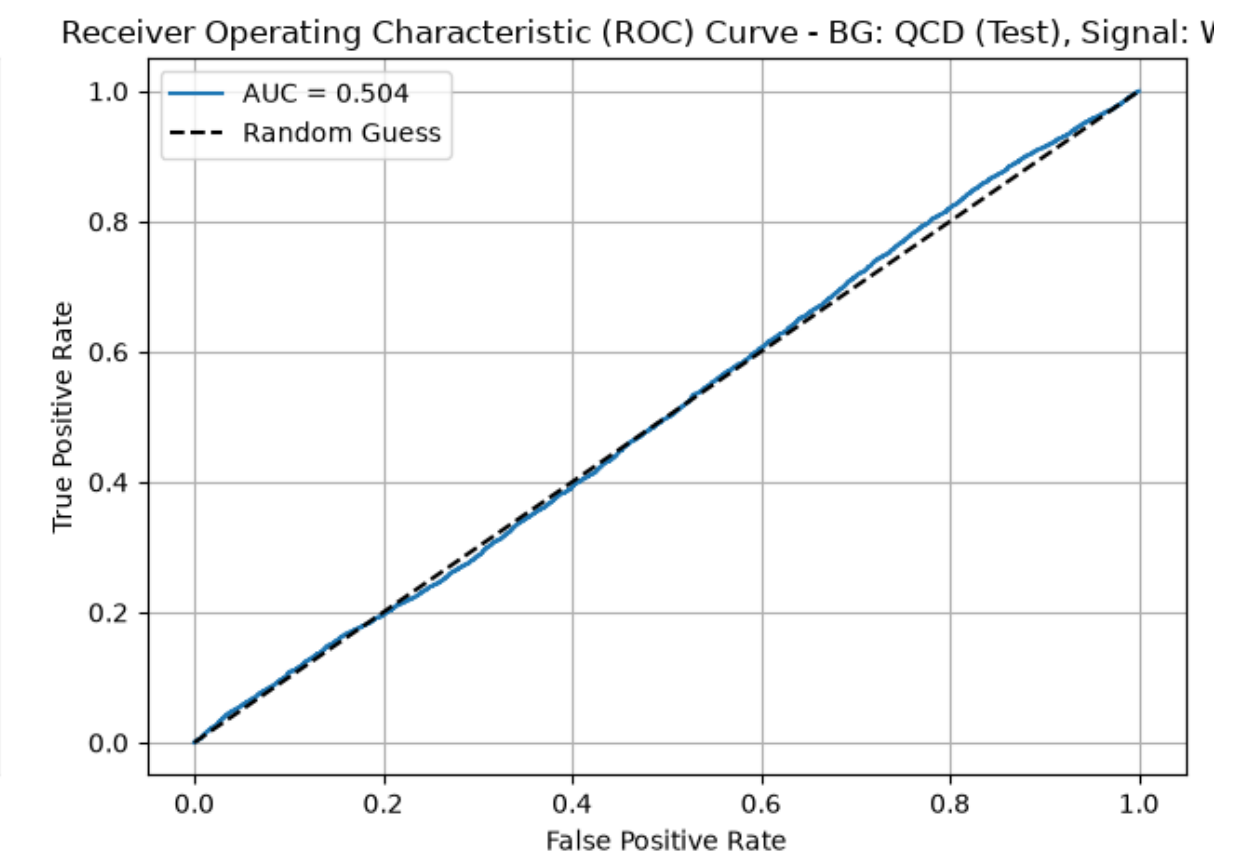
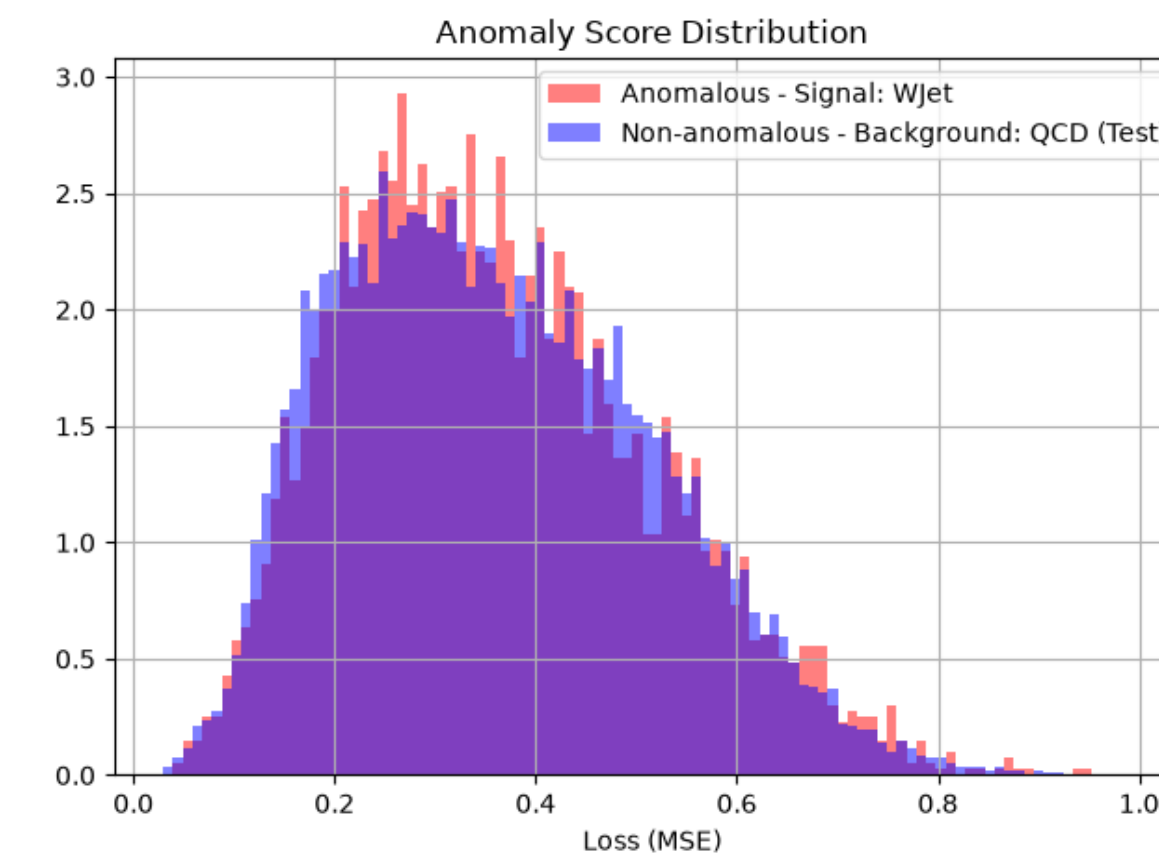
# CLS-conditioned Prototype Reconstruction

## ROC Performance Volatile Depending on Hyper-params

It is worth noting that it follows the same trend that as the capacity grows, the model generalizes to unseen anomalous samples, thus reducing AUC.

However, contrasting to GAE, as training continues, Transformer AE reconstructs WJet (unseen signal) events increasingly well, and the ROC AUC drops toward 0.5.

Top plots: Transformer w/ 32 hidden, 8 latent, 4 layers, 150 epochs  
Bottom plots: same model, but at 20th epoch





# Position-conditioned Token-to-Token Reconstruction

## Constraining the Decoder to Avoid Template Fitting

The CLS-conditioned prototype model enforces event-level compression, but its decoder is highly flexible: it starts from learned slots and uses Hungarian matching to choose the best assignment to target particles. This can turn reconstruction into template fitting, where the decoder learns a general particle-set generator rather than a strict event reconstructor. To reduce this freedom, we keep the CLS bottleneck but replace prototype decoding with position-conditioned token-to-token reconstruction.

The encoder is the same as before, but the decoder no longer starts from learned output slots. Instead, for each input particle, the decoder receives that particle's eta-phi position encoding added by the shared CLS latent context. It then reconstructs the particle features at the corresponding token position.

The training target is token-wise reconstruction over all particle tokens. Since each decoder output is tied to a specific input particle position, no Hungarian matching is needed. The loss is the MSE between reconstructed and target particle features, averaged over particle tokens. At evaluation time, the event-averaged reconstruction loss is used as the anomaly score.

# Position-conditioned Token-to-Token Reconstruction

## Results

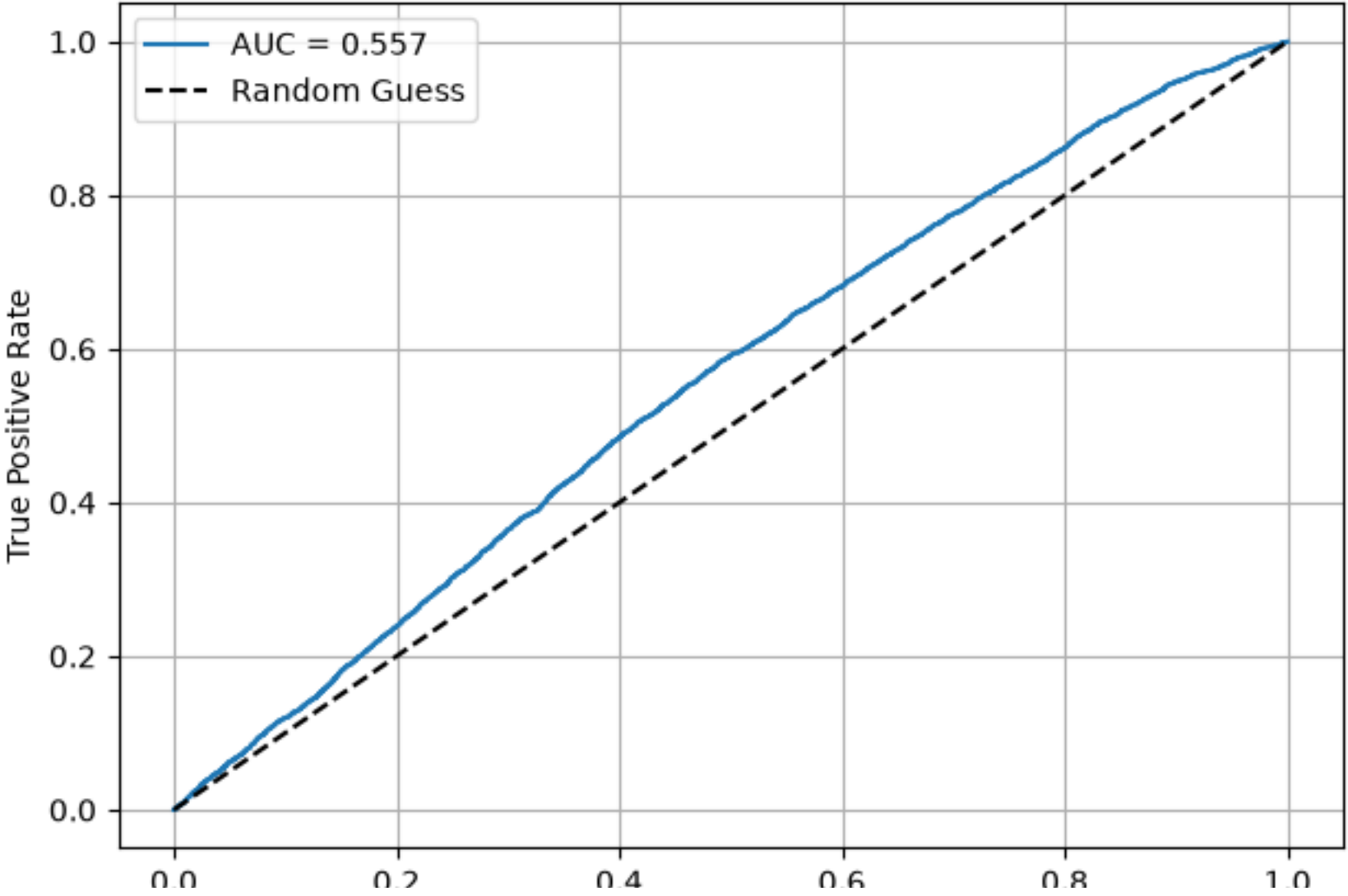
Across different model sizes, the position-conditioned token-to-token model gives poor ROC performance, close to random guessing and consistently worse than the prototype reconstruction model.

This suggests that once each decoder output is tied to the corresponding particle’s position, the reconstruction task becomes too local and too easy to generalize across QCD and WJet. Given the global CLS context and a particle’s position, the rule for reconstructing its features appears to be largely shared by the two event distributions.

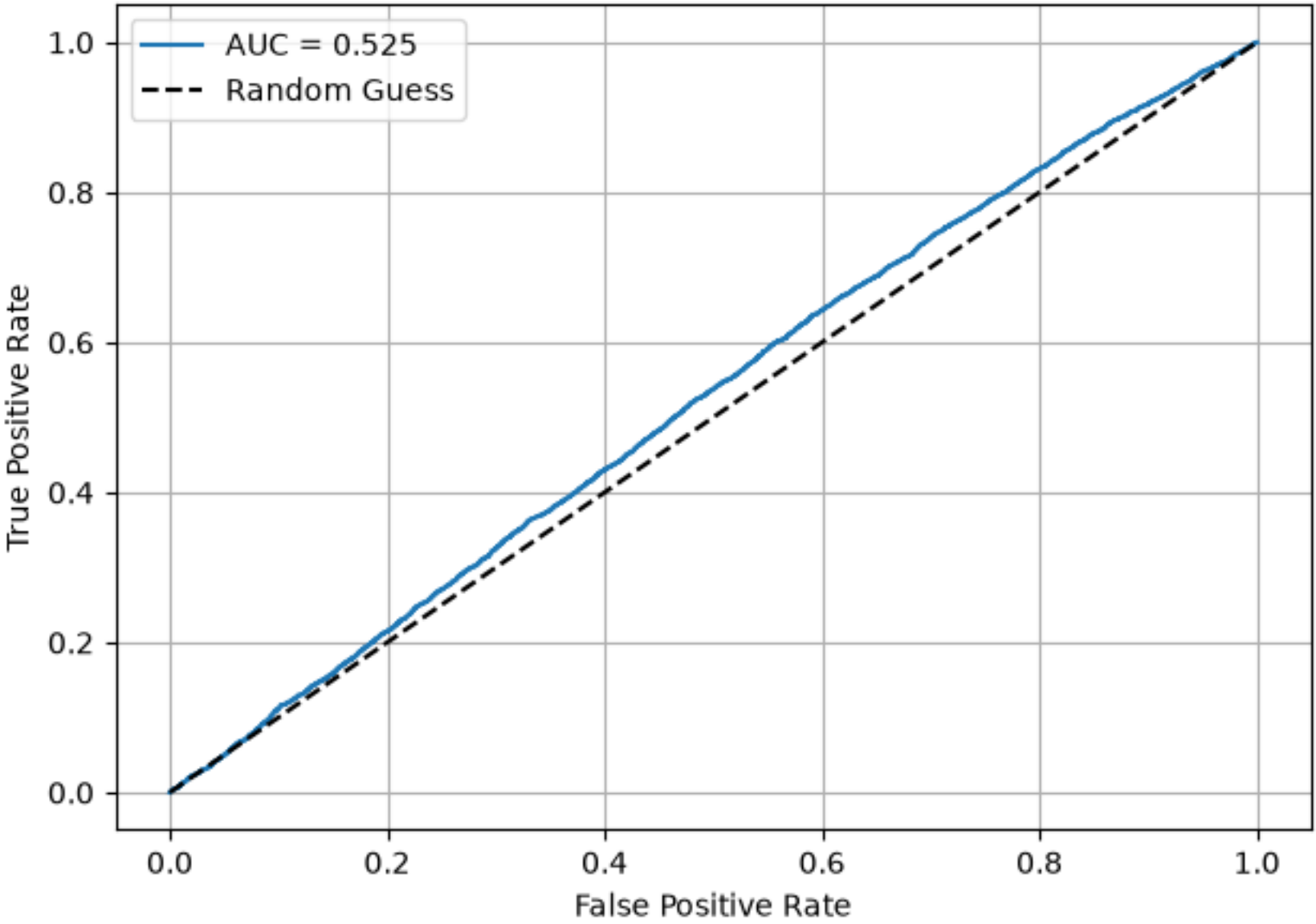
Top: ROC for 4 hidden, 2 latent, 4 layers

Bottom: ROC for 64 hidden, 16 latent, 4 layers

Receiver Operating Characteristic (ROC) Curve - BG: QCD (Test), Signal: V



Receiver Operating Characteristic (ROC) Curve - BG: QCD (Test), Signal: V

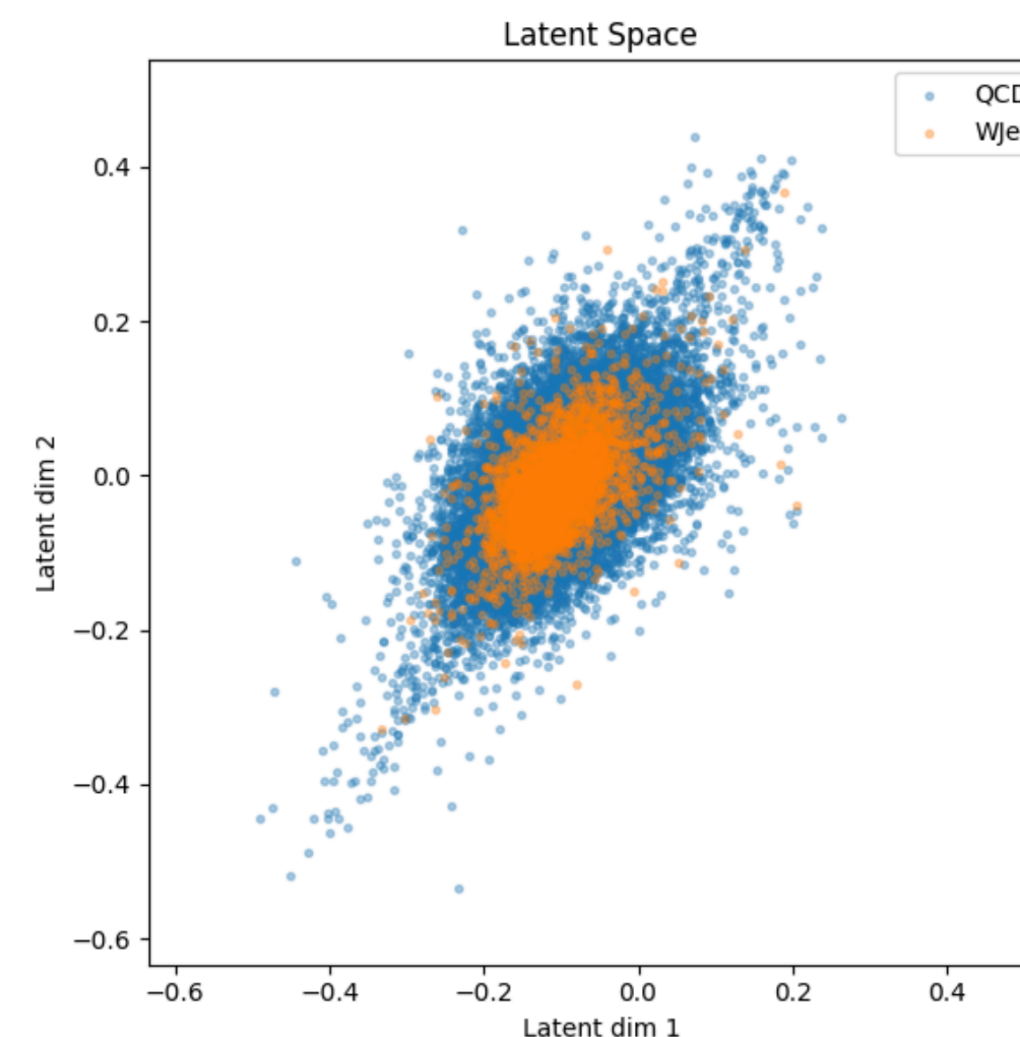
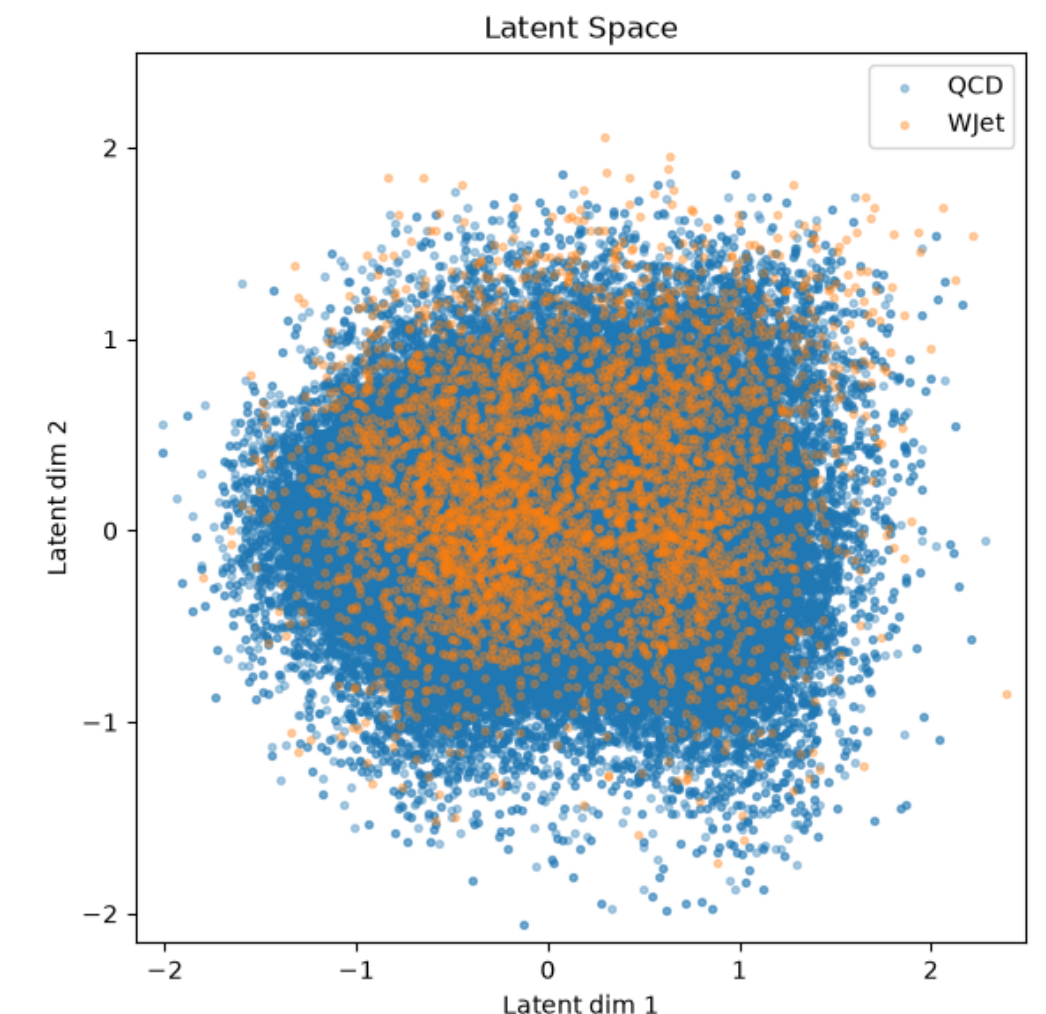
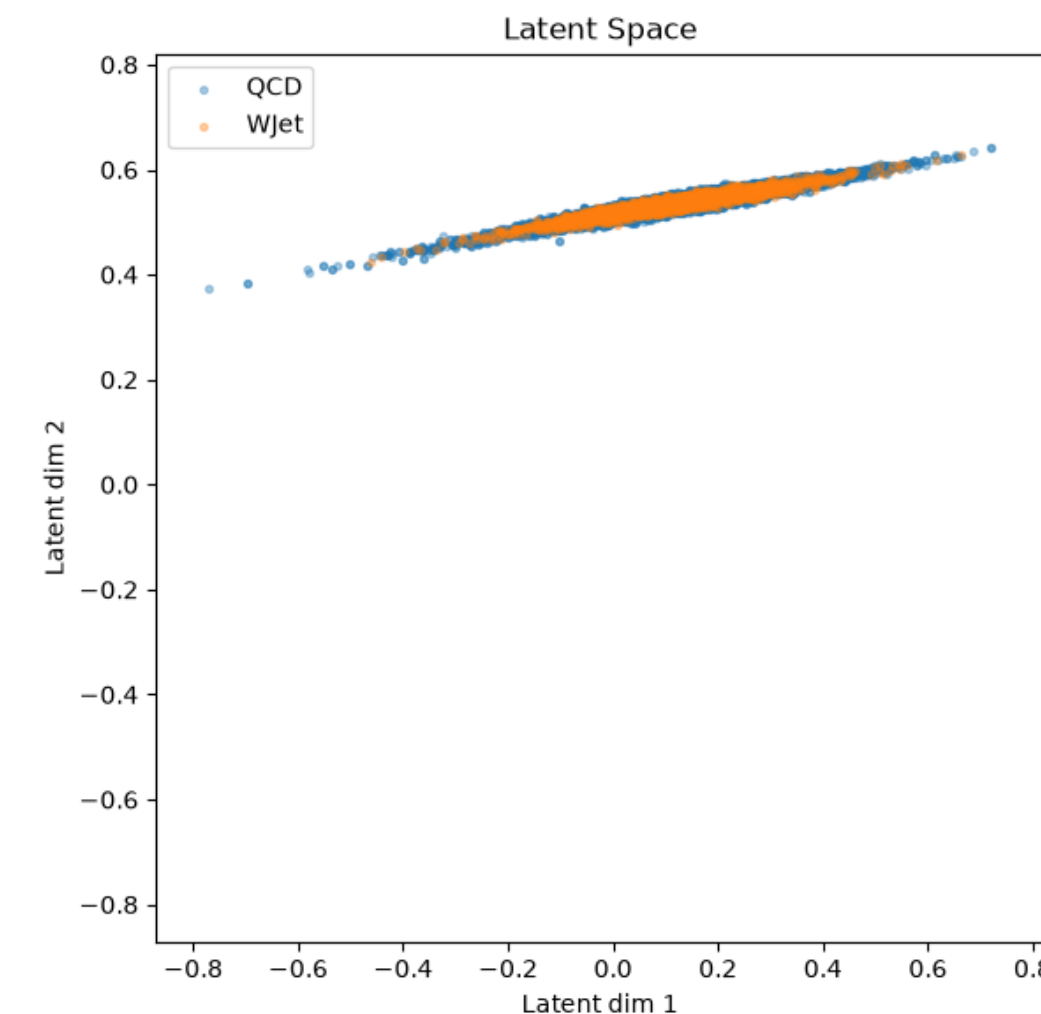


# Plotting Latent-Space

## Reconstruction-based Anomaly Detection Is A Decoder Job

We also visualized latent spaces for both GAE and Transformer autoencoders.

In most cases, QCD and WJet overlap strongly in the latent space. This means the encoder itself is not learning a representation that directly separates the two distributions. The anomaly score is therefore mainly determined by the decoder residual: whether the decoder reconstructs WJet slightly worse than QCD. This makes the method fragile, because a more expressive decoder can easily remove the residual gap without making the latent representation more physically meaningful.



Top left: Prototype Transformer AE w/ 4 hidden, 2 latent, 4 layers

Top right: Prototype Transformer AE w/ 32 hidden, 8 latent, 4 layers

Bottom left: GAE w/ 8 hidden, 2 latent, 3 kNN



# Future Direction

## From Reconstruction-based to Representation-based

Overall, these results suggest that reconstruction loss alone is not a reliable long-term objective for anomaly detection.

The current AE-based methods can work when the model is sufficiently constrained, but their success depends on capacity limitation and decoder failure rather than on a representation that explicitly separates normal and anomalous events. If confronting a completely unknown anomaly, we cannot reliably determine capacity sweet spot or human-factor feature engineering bias and hope a discerning reconstruction loss arises as the byproduct.

A more principled future direction is to reformulate the problem as representation learning: instead of hoping that WJet produces a larger reconstruction error, we want the latent space itself to encode distribution-level differences between QCD and WJet, and eventually to support more robust detection of unknown anomalies. This requires physics semantic preserving and distorting data augmentations to generate synthetic positive and negative samples for contrastive learning or joint embedding learning.